

Portfolio Website

Complete Project Guide — Code + Design Explanation

HTML5 · CSS3 · Bootstrap 5 · JavaScript · Resume-Ready

Project	Personal Portfolio Website — Single Page Application
Sections	Navbar · Hero · About · Skills · Projects · Contact · Footer
Technologies	HTML5, CSS3, Bootstrap 5, Vanilla JavaScript
Key Features	Typed text animation, Scroll animations, Form validation, Responsive
Purpose	Resume showcase project — impress employers with frontend skills
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Year	2026

Section 1 — Project Structure & Architecture

1.1 File Structure

The entire portfolio is ONE HTML file (index.html). CSS is in a separate style.css, and all JavaScript in main.js. This clean separation is what interviewers look for in a junior developer.

```
portfolio/
■■■ index.html      ← One page, all sections (navbar to footer)
■■■ css/
■   ■■■ style.css   ← Custom styles + CSS variables
■■■ js/
■   ■■■ main.js     ← All JS: navbar, typed text, animations, form
■■■ images/         ← Your photo, project screenshots (optional)

In index.html <head>:
  <link href="bootstrap.min.css" rel="stylesheet">    ← Bootstrap first
  <link rel="stylesheet" href="css/style.css">         ← Your CSS second

Before </body>:
  <script src="bootstrap.bundle.min.js"></script>    ← Bootstrap JS
  <script src="js/main.js"></script>                 ← Your JS last
```

1.2 Skills This Project Demonstrates to Employers

Technical Skill	Where in Code	Why Employers Care
Semantic HTML5	All sections use proper tags	SEO-aware, accessible code
CSS Custom Properties	:root variables in style.css	Maintainable, modern CSS
Bootstrap 5 Grid	Skills, Projects, Contact rows	Industry standard framework
CSS Animations	Hero float, fade-in sections	UI polish & attention to detail
Intersection Observer API	Skill bars + section animations	Performance-aware JS
Typed text JS effect	Hero section typing animation	DOM manipulation skill
Form Validation	Contact form novalidate + JS	UX + data handling
Responsive Design	All sections, col-md-6 col-lg-4	Mobile-first mindset
Git / GitHub Pages	Hosting the live demo	Dev workflow knowledge

Section 2 — HTML Structure Walkthrough

2.1 Navbar — Fixed, Responsive, Scroll-aware

The navbar uses Bootstrap's navbar-expand-lg to be horizontal on desktop and collapse to a hamburger on mobile. fixed-top keeps it visible during scroll. JavaScript adds a scrolled CSS class when user scrolls 50px down.

```
<nav class="navbar navbar-expand-lg navbar-dark fixed-top" id="mainNav">
  <div class="container">

    <!-- Logo / Brand (your initials in code style) -->
    <a class="navbar-brand fw-bold fs-4" href="#home">
      <span class="text-warning">&lt;</span>YourName<span class="text-warning">&gt;></span>
    </a>

    <!-- Hamburger (auto shown on mobile by Bootstrap) -->
    <button class="navbar-toggler border-0" type="button"
      data-bs-toggle="collapse" data-bs-target="#navLinks">
      <span class="navbar-toggler-icon"></span>
    </button>

    <!-- Nav Links -->
    <div class="collapse navbar-collapse" id="navLinks">
      <ul class="navbar-nav ms-auto gap-1">
        <!-- ms-auto pushes links to RIGHT -->
        <li class="nav-item"><a class="nav-link" href="#home">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#about">About</a></li>
        <li class="nav-item"><a class="nav-link" href="#skills">Skills</a></li>
        <li class="nav-item"><a class="nav-link" href="#projects">Projects</a></li>
        <li class="nav-item"><a class="nav-link" href="#contact">Contact</a></li>
      </ul>
      <!-- Resume download button -->
      <a href="resume.pdf" download class="btn btn-warning btn-sm ms-3 fw-bold">
        &#128190; Resume
      </a>
    </div>
  </div>
</nav>
```

Note: The `data-bs-target='#navLinks'` must match `id='navLinks'` exactly. Bootstrap JS uses this to know which element to collapse/expand.

2.2 Hero Section — Full-Screen First Impression

The hero section must make an impact in 3 seconds. The row uses `order-2 order-lg-1` so on mobile the text appears below the avatar, but on desktop the text is on the left.

```

<section class="hero-section d-flex align-items-center" id="home">
  <div class="container">
    <div class="row align-items-center min-vh-100 py-5">

      <!-- TEXT — order-2 on mobile (below avatar), order-lg-1 on desktop (left) -->
      <div class="col-lg-7 order-2 order-lg-1">
        <p class="greeting mb-2">&#128075; Hello, I'm</p>
        <h1 class="display-3 fw-bold mb-2">Your Full Name</h1>

        <!-- This text is animated by JS: types, deletes, repeats -->
        <h2 class="typed-text mb-4">Frontend Developer</h2>

        <p class="lead text-white-75 mb-4" style="max-width:520px;">
          A short, powerful 2-line description of what you do and where you trained.
          Trained at ForkInfosystems, Karad.
        </p>

        <!-- Two CTA buttons -->
        <div class="d-flex gap-3 flex-wrap mb-5">
          <a href="#projects" class="btn btn-warning btn-lg fw-bold px-4">
            View My Work &#8594;
          </a>
          <a href="#contact" class="btn btn-outline-light btn-lg px-4">
            Hire Me
          </a>
        </div>
      </div>

      <!-- AVATAR — order-1 on mobile (above text), order-lg-2 (right) -->
      <div class="col-lg-5 order-1 order-lg-2 text-center mb-5 mb-lg-0">
        <!-- Floats up/down via CSS @keyframes float animation -->
        <div class="hero-avatar mx-auto">
          <span>&#128104;&#8205;&#128187;</span>
        </div>
        <!-- Tech badges below avatar -->
        <div class="d-flex flex-wrap justify-content-center gap-2 mt-4">
          <span class="badge bg-warning text-dark px-3 py-2">HTML5</span>
          <span class="badge bg-primary px-3 py-2">CSS3</span>
          <span class="badge bg-success px-3 py-2">Bootstrap</span>
          <span class="badge bg-danger px-3 py-2">JavaScript</span>
        </div>
      </div>
    </div>
  </div>
</section>

```

2.3 Projects Section — Portfolio Cards

Project cards are the MOST important part of your portfolio. Each card needs: project name, description (2-3 lines), tech stack tags, Live Demo link, and GitHub link.

```

<!-- One project card – copy and repeat for each project -->
<div class="col-md-6 col-lg-4">
  <div class="project-card h-100">

    <!-- Gradient header – use different colours per project -->
    <div class="project-img"
      style="background: linear-gradient(135deg,#1a237e,#e65100);">
      <span>&#128722;</span> <!-- emoji or <img> screenshot -->
    </div>

    <div class="project-body p-4">
      <div class="d-flex justify-content-between align-items-start mb-2">
        <h5 class="fw-bold mb-0">ShopZone E-Commerce</h5>
        <span class="badge bg-primary">Major</span> <!-- project type badge -->
      </div>
      <p class="text-muted small mb-3">
        Full e-commerce UI with product grid, shopping cart, admin panel,
        and login/register. Fully responsive with Bootstrap 5.
      </p>
      <!-- Tech stack pills -->
      <div class="d-flex flex-wrap gap-1 mb-3">
        <span class="tech-tag">HTML</span>
        <span class="tech-tag">Bootstrap</span>
        <span class="tech-tag">JavaScript</span>
      </div>
      <!-- Links: replace # with real URLs -->
      <div class="d-flex gap-2">
        <a href="https://yourusername.github.io/shopzone"
          target="_blank" class="btn btn-sm btn-primary flex-fill">
          &#128065; Live Demo
        </a>
        <a href="https://github.com/yourusername/shopzone"
          target="_blank" class="btn btn-sm btn-outline-dark flex-fill">
          &#128196; GitHub
        </a>
      </div>
    </div>
  </div>
</div>

```

Section 3 — CSS Design System Explained

3.1 CSS Variables — The Whole Theme in 5 Lines

Define all colours and values once in `:root`. Changing `--primary` changes the entire site instantly. This is what senior developers call 'maintainable CSS'.

```
/* css/style.css — Top of file — CHANGE THESE TO PERSONALISE */
:root {
  --primary:      #1a237e; /* ← Your main colour (navbar, headings, buttons) */
  --accent:       #e65100; /* ← Accent colour (hover effects, underlines) */
  --dark:         #0d1117; /* ← Dark background (hero, footer) */
  --card-shadow:  0 4px 20px rgba(0,0,0,0.08);
  --hover-shadow: 0 12px 35px rgba(0,0,0,0.15);
  --transition:   all 0.3s ease;
}

/* Usage examples */
.project-card { box-shadow: var(--card-shadow); }
.project-card:hover {
  box-shadow: var(--hover-shadow);
  transform: translateY(-6px);
  transition: var(--transition);
}
.section-title::after {
  content: ''; display: block;
  width: 55px; height: 4px;
  background: var(--accent); /* accent underline on section titles */
  margin: 10px auto 0;
}
```

3.2 Hero Gradient + Float Animation

```
/* Hero gradient background */
.hero-section {
  min-height: 100vh;
  background: linear-gradient(
    135deg,
    #0d1117 0%,      /* dark start */
    #1a237e 50%,     /* navy middle */
    #0d1117 100%     /* dark end */
  );
  position: relative;
  overflow: hidden;
}

/* Subtle orange glow using pseudo-element */
.hero-section:before {
  content: '';
  position: absolute; inset: 0;
  background: radial-gradient(
    ellipse at 60% 50%,
    rgba(230,81,0,0.15) 0%, transparent 60%
  );
}

/* Avatar floats up and down continuously */
.hero-avatar {
  width: 220px; height: 220px;
  border-radius: 50%;
  animation: float 4s ease-in-out infinite;
}

@keyframes float {
  0%, 100% { transform: translateY(0px); }
  50%      { transform: translateY(-15px); }
```

3.3 Scroll Fade-In Animation Pattern

```
/* Step 1: CSS – starting state (invisible, shifted down) */
.fade-in-up {
  opacity: 0;
  transform: translateY(30px);
  transition: opacity 0.6s ease, transform 0.6s ease;
}
/* When JS adds 'visible' class → animate to final state */
.fade-in-up.visible {
  opacity: 1;
  transform: translateY(0);
}

/* Step 2: Add class to elements in HTML */
<div class="card fade-in-up">...</div>
<section class="fade-in-up">...</section>

/* Step 3: JS – use IntersectionObserver (see JS section) */
/* When element enters viewport → add 'visible' class */
/* Much better than animating all at once on page load! */
```

Section 4 — JavaScript Functionality Explained

4.1 Navbar Scroll Effect

```
// js/main.js

// 1. Navbar scroll effect – add class when user scrolls down
window.addEventListener('scroll', () => {
  const nav = document.getElementById('mainNav');
  // classList.toggle(class, force) – adds if force=true, removes if false
  nav.classList.toggle('scrolled', window.scrollY > 50);
});

// In CSS:
// #mainNav { background: rgba(13,17,23,0.95); backdrop-filter: blur(12px); }
// #mainNav.scrolled { box-shadow: 0 4px 20px rgba(0,0,0,0.3); }
```

4.2 Typed Text Cycling Animation

```
// 2. Typed text effect – cycles through job titles
const roles = [
  'Frontend Developer',
  'UI/UX Designer',
  'JavaScript Developer',
  'Bootstrap Expert'
];

let roleIdx = 0, charIdx = 0, isDeleting = false;
const typedEl = document.querySelector('.typed-text');

function typeRole() {
  const current = roles[roleIdx];
  if (!isDeleting) {
    typedEl.textContent = current.slice(0, ++charIdx);
    if (charIdx === current.length) {
      isDeleting = true;
      setTimeout(typeRole, 1800); // pause at full word
      return;
    }
  } else {
    typedEl.textContent = current.slice(0, --charIdx);
    if (charIdx === 0) {
      isDeleting = false;
      roleIdx = (roleIdx + 1) % roles.length;
    }
  }
  setTimeout(typeRole, isDeleting ? 55 : 95);
}

typeRole(); // kick off the animation
```

4.3 Intersection Observer — Animate on Scroll

```
// 3. Animate elements when they scroll into view
const observer = new IntersectionObserver((entries) => {
  entries.forEach(entry => {
    if (entry.isIntersecting) {
      entry.target.classList.add('visible'); // triggers CSS transition
      observer.unobserve(entry.target);      // only animate once
    }
  });
}, { threshold: 0.15 }); // fire when 15% visible

// Apply to all fade-in elements
document.querySelectorAll('.fade-in-up').forEach(el => observer.observe(el));

// 4. Animate skill bars when skills section comes into view
const skillObserver = new IntersectionObserver((entries) => {
  entries.forEach(entry => {
    if (entry.isIntersecting) {
      entry.target.querySelectorAll('.progress-bar').forEach(bar => {
        const targetW = bar.style.width; // save the target %
        bar.style.width = '0%';          // reset to 0
        setTimeout(() => bar.style.width = targetW, 200); // animate!
      });
      skillObserver.unobserve(entry.target);
    }
  });
}, { threshold: 0.3 });
skillObserver.observe(document.getElementById('skills'));
```

4.4 Contact Form Validation + UX

```
// 5. Contact form with Bootstrap validation + loading state
const form = document.getElementById('contactForm');
form.addEventListener('submit', function(e) {
  e.preventDefault(); // prevent page reload

  if (!this.checkValidity()) {
    this.classList.add('was-validated'); // Bootstrap shows error styles
    return;
  }

  const btn = this.querySelector('button[type="submit"]');

  // Show loading spinner in button
  btn.innerHTML = '<span class="spinner-border spinner-border-sm me-2">'
    + '</span>Sending...';
  btn.disabled = true;

  // Simulate API call – replace with real fetch() to EmailJS or backend
  setTimeout(() => {
    btn.innerHTML = '&#10003; Message Sent!';
    btn.className = 'btn btn-success w-100 py-3 fw-bold';
    this.reset();
    this.classList.remove('was-validated');
    // Restore button after 3 seconds
    setTimeout(() => {
      btn.innerHTML = 'Send Message &#10148;';
      btn.className = 'btn btn-primary w-100 py-3 fw-bold';
      btn.disabled = false;
    }, 3000);
  }, 1500);
});
```

Section 5 — Personalise, Deploy & Interview Tips

5.1 Personalisation Steps

1 Change name, role, and greeting text

- index.html → Hero section → Find 'Aarav Sharma' → replace with your full name.
- main.js → const roles array → update with YOUR actual job titles.
- Navbar brand → change the abbreviation to your initials.

2 Update About section with your details

- About section → Change education (BCA → your degree), city, email, phone.
- Change the bio paragraph to your real background.
- Change 'ForkInfosystems' in bio to your institute name.

3 Adjust skill percentages honestly

- Skills section → Each progress bar has style='width:XX%' → update to your level.
- 90% HTML is fine. 50-60% JS shows honest self-awareness.
- Add/remove skill bars to match your actual skills.

4 Replace projects with your real work

- Replace placeholder text in each .project-card with your real projects.
- Change href='#' in Live Demo → your hosted URL.
- Change href='#' in GitHub → your actual GitHub repo URL.

5 Add your real contact details

- Contact section → update email, phone, LinkedIn, GitHub, city.
- Footer → update name and training info.

6 Add your actual resume PDF

- Place your resume file as 'resume.pdf' in the root portfolio folder.
- The Resume button already has download attribute — no code change needed.

7 Host free on GitHub Pages

- Push portfolio folder to GitHub. Settings → Pages → Deploy from main.
- Live at: <https://yourusername.github.io/portfolio/>
- Add this URL to your resume and LinkedIn profile!

5.2 GitHub Pages Deployment

```
# Open Terminal / Git Bash in your portfolio folder

# Step 1: Initialise git
git init

# Step 2: Stage all files
git add .

# Step 3: First commit
git commit -m "Portfolio website - initial commit"

# Step 4: Create repo on github.com first, then:
git remote add origin https://github.com/YOURUSERNAME/portfolio.git
git branch -M main
git push -u origin main

# Step 5: Enable GitHub Pages
# github.com → Your repo → Settings → Pages
# Source: Deploy from branch → main → /(root) → Save

# Step 6: Wait 2 minutes → your site is live at:
# https://YOURUSERNAME.github.io/portfolio/

# After any changes:
git add .
git commit -m "Update projects section"
git push
```

5.3 Interview Talking Points

Question	Key Points to Say	Shows
Tell me about this project	Single-page responsive portfolio using HTML5, Bootstrap 5, Vanilla JS. Features scroll animations using Intersection Observer API, typed text effect, Bootstrap form validation, and is deployed on GitHub Pages.	Technical breadth + communication
Why Bootstrap over pure CSS?	Bootstrap speeds up development with its responsive 12-column grid and pre-built, tested components. I layered CSS custom properties on top for branding.	Framework knowledge + customisation
How did you make it responsive?	Mobile-first Bootstrap grid: col-12 for mobile, col-md-6 for tablet, col-lg-4 for desktop. CSS media queries handle typography and spacing.	Responsive design depth
What is the hardest part?	IntersectionObserver API for scroll animations — understanding async callbacks and when elements enter the viewport took research.	Problem-solving honesty
What would you add next?	EmailJS for real contact form sending, React to make it component-based, and a blog section using a headless CMS like Contentful.	Growth mindset, next steps